Date:
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Study of various network commands used in Linux and windows:

AIM:

To study of various Networking commands used in Linux and windows.

i) **Command:**

arp -a arp is short form of address resolution protocal It will show the ip address of your computer along with the IP address and MAC adress of your router.

output:

```
172.16.9.212    at  7C:57:58:38:da:a3  ether on enp380
172.16.8.186    at  7C:57:58:38:e6:24  ether on enp380.
172.16.8.249    at  00:27:0e:13:f5:77  ether on enp380.
172.16.11.207   at  7C:57:58:38:46:87  ether on enp380.
```

ii) **command:** hostname

This is the simplest of all TCP/IP commands. It simply display the name of the computer.

output:

localhost.localdomain.

iii) ~~Ipconfig~~ **command:** ipConfig /all

This command displays detailed information about your TCP/IP connection.

output:

```
DNS server : 172.16.8.1
NetBIOS over Tcpip : Enabled.
Subnet mask . : 255.255.252.0
```

iv) nbtstat : netset displays a variety of statistics about a computers active TCP/IP connections. It is a command like tool for monitoring network connections both incoming and outgoing.

v) netstat: displays a variety of stastics about computer active TCP/IP connections.

output: netstat -r

Destination	Gateway	Genmask	Flags	Mss	Window	irtt	Iface
default	gateway	0.0.0.0	UG	0	0	0	eth0
172.16.8.0	0.0.0.0	255.255.252.	U	0	0	0	enp3s0

vi) Ping: (packet Internet groper) is the best way to test connectivity between two nodes. ping use ICMP to

output: ping google.com.

64 bytes from pnmaaa-aq-in-f14.1e100.net (142.250.77.110): icmp_seq=1 ttl=118 time 21.8 ms.

vii) pathping: Pathping is unique to window's and basically a combination of ping and Tracert commands.

pathping google.com

output:

Trace route to google.com [142.250.77.110]

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Linux commands.

1. ip address show : IP command show address information.

output:

1: lo: <LoopBack, up, Lower-up> mtu 65536 ..

inet 127.0.0.1/8

2. If config : ifconfig was a staple in many sysadmin's tool belt for configuration and Networking trouble shooting.

output:

enp 2s0:

inet 172.16.9.800

netmask 255.255.252.0

broadcast 172.16.11.255

3. mtr : used to analyze the path packets take from your computer to destination. measure qty of each hop along route.

output:

fedora(::1) -> localhost(::1)

Packets.

loss%	snt	last	avg	Best	wrst	stdev
0.0	71	0.1	0.1	0.0	0.1	0.0

4. tcpdump : tcp command is designed to r capturing and displaying packets.

5. ping google.com

output: 142.250.77.110 56(84) bytes of data

Result:

Study of Basic Linux commands are succesfully runned and verifyed.